

ASSIGNMENT 3CHE - 321

Problem 1

Ex 4-1

Yes, the worked example ~~is~~ will still be correct if water was considered inert, under the assumption

- ① Liquid phase, constant density $\Rightarrow V = V_0$
- ② Water does not participate in any reaction or evaporate / absorb
- ③ No significant volume change on mixing

the presence of inert only dilutes the reacting species, it does not change the concentration-conversion relations:

$$C_A = C_{A0}(1-X), \quad C_B = C_{A0}\left(\Theta_B - \frac{1}{3}X\right), \quad C_C = C_{A0}(\Theta_C + X)$$

$$C_D = C_{A0}\left(\Theta_D + \frac{1}{3}X\right)$$

If the water were not inert (reacted, volatilized etc.) the concentration would have changed & the expressions need to be modified.

Eg 4-2.

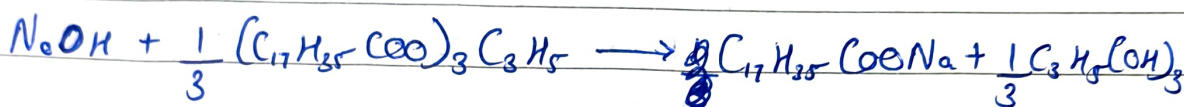


$$C_{A0} = 10 \text{ mol/dm}^3 \text{ (NaOH)}$$

$$C_{B0} = 3 \text{ mol/dm}^3 \text{ (glyceryl stearate)}$$

$$C_{C0} = C_{D0} = 0$$

We want $X = 20\%$ & $X = 90\%$



$$\Theta_B = 0.3, \quad \Theta_C = 0, \quad \Theta_D = 0$$

$$C_A = C_{A0}(1-X) = 10(1-0.20) = 8 \text{ mol/dm}^3$$

$$C_B = C_{B0}(\Theta_B - \frac{X}{3}) = 10(0.3 - \frac{0.2}{3}) = 2.33 \text{ mol/dm}^3$$

$$C_C = C_{A0}(\Theta_C + X) = 10(0.20) = 2 \text{ mol/dm}^3$$

$$C_D = C_{A0}(\Theta_D + X) = 10(\frac{0.20}{3}) = 0.67 \text{ mol/dm}^3$$

$$\underline{\underline{X = 90}}$$

$$C_A = 10(1-0.9) = 1 \text{ mol/dm}^3$$

$$C_B = 10(0.3 - \frac{0.9}{3}) = 0 \text{ mol/dm}^3$$

$$C_C = 10(0.9) = 9 \text{ mol/dm}^3$$

$$C_D = 10(\frac{0.90}{3}) = 3 \text{ mol/dm}^3$$

Eg 4-3

We have,

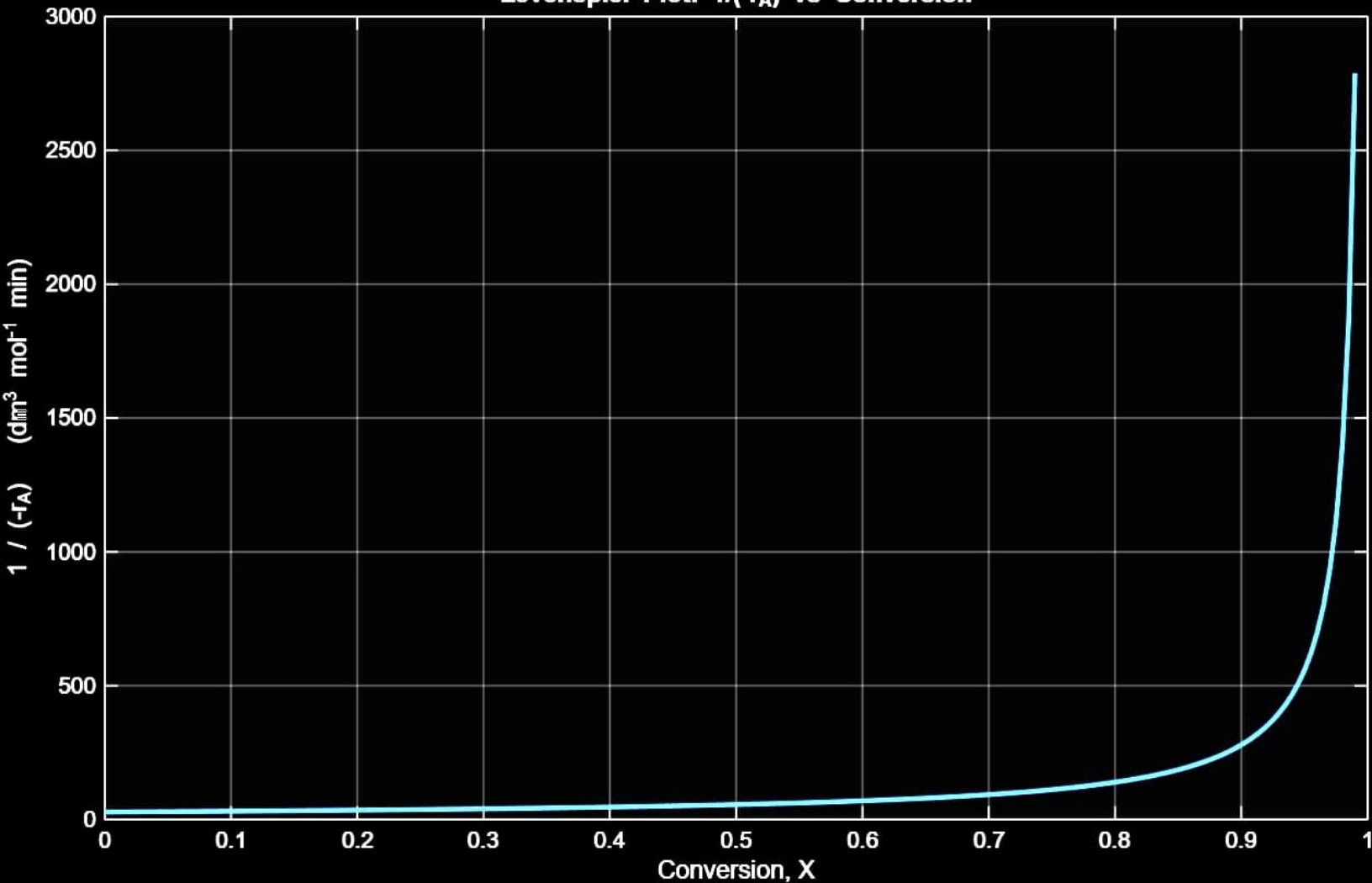
$$C_1(x) = \frac{C_{A0} \Theta_1}{1 + \epsilon X} \frac{P}{P_0} \frac{T_0}{T}$$

With the above eq. into consideration we can say that $C_1(x)$ (concentration of N_2) will remain constant if the factor

$$\boxed{\frac{P/T}{1 + \epsilon X}} \rightarrow \text{is constant}$$

Another case might be when the reaction is isothermal & has no pressure drop $\epsilon = 0$, hence no change in moles $\rightarrow$ a constant conc. of N_2

Levenspiel Plot: $1/(-r_A)$ vs Conversion



Eg 4-4

We have

$$K_c = \frac{C_B^2}{C_A}$$

$$\text{PFR} \rightarrow C_A = C_{A0} \frac{1-X}{1+EX}, \quad C_B = C_{A0} \frac{2X}{1+EX}$$

$$\text{Hence } K_c = \frac{4 C_{A0}^2 X^2}{(1+EX)(1-X)} \quad \text{--- (1)}$$

→ batch

$$C_A = C_{A0} (1-X), \quad C_B = 2 C_{A0} X$$

$$K_c = \frac{4 C_{A0}^2 X^2}{1-X} \quad \text{--- (2)}$$

① has an extra $(1+EX)$ in the denominator which means

$$X_{e, \text{PFR}} > X_{e, \text{batch}}$$

to keep K_c the same

↳ No it's not always same

$\Delta V = 0$ ($\xi = 0$) both are identical

$\Delta V < 0$, $(1+EX) < 1$, so PFR gives the lower equilibrium conversion.

For ~~Batch~~ Batch

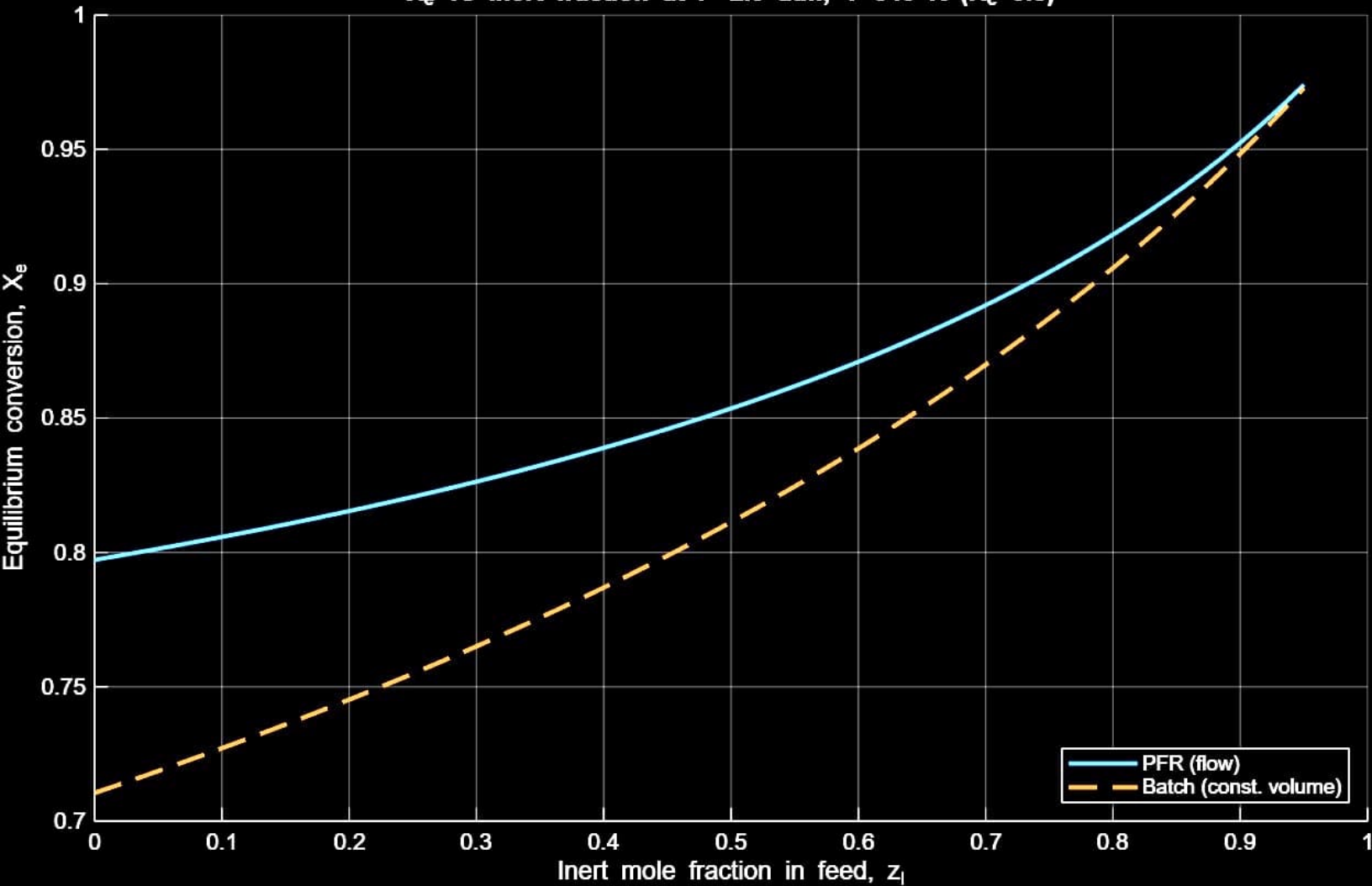
$$X_{eb} = \frac{-K_c + \sqrt{K_c^2 + 16C_{A0}K_c}}{8C_{A0}} \rightarrow \text{using } \textcircled{2}$$

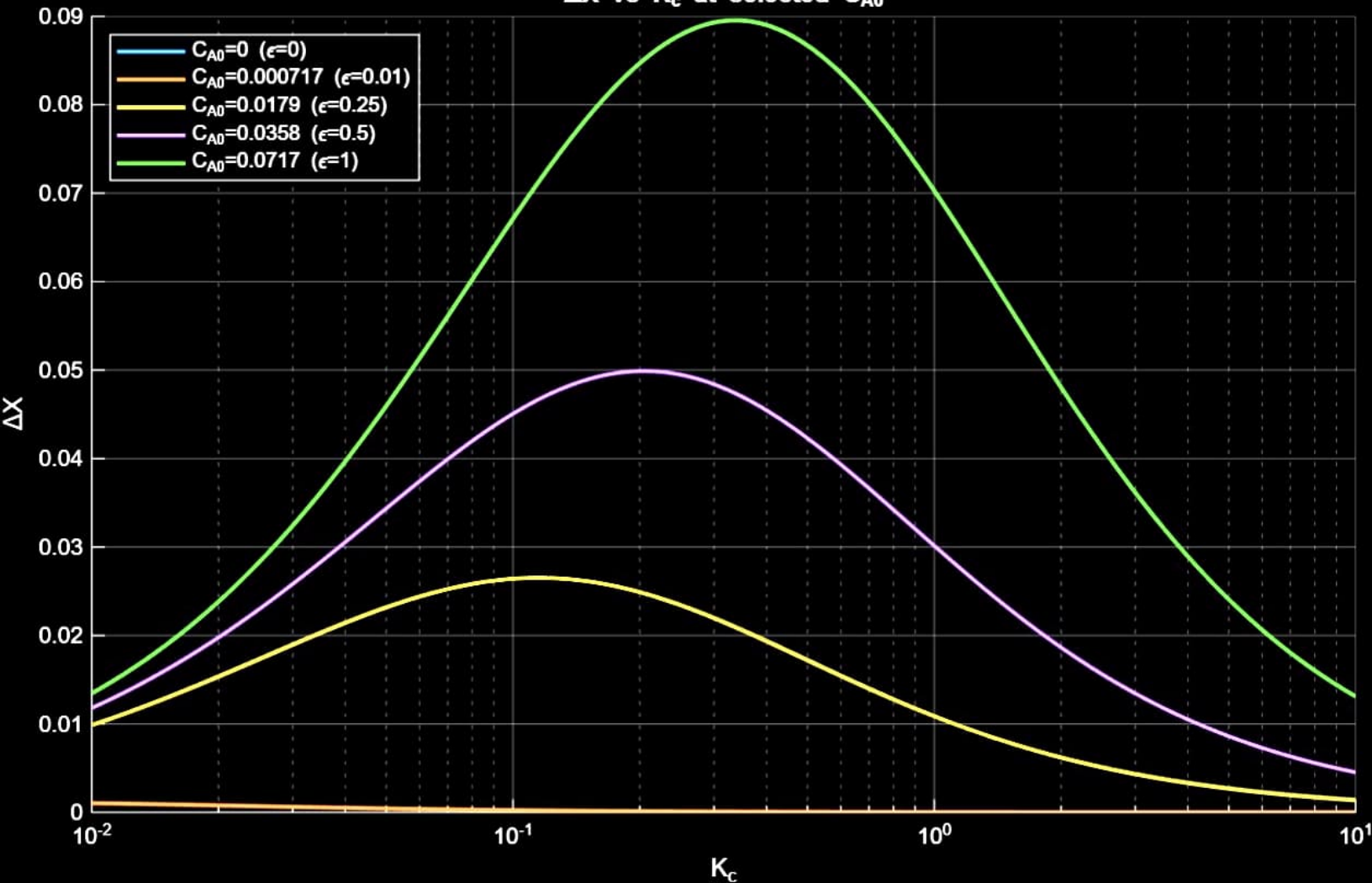
For PFR

$$X_{eb} = \frac{K_c(C_A - 1) + \sqrt{K_c^2(C_A - 1)^2 + 4(K_c C_A + 4C_{A0})K_c}}{2(K_c C_A + 4C_{A0})}$$

→ using $\textcircled{1}$

X_e vs inert fraction at $P=2.0$ atm, $T=340$ K ($K_c=0.5$)



ΔX vs K_c at selected C_{A0} 

Eg 4.5

②

$X = 0.40$

From the example we have

$$-r_A(X) = 0.036 \left[\frac{1-X}{1+X} - \frac{2.88X^2}{(1+X)^2} \right] \left(\frac{\text{mol}}{\text{dm}^3 \cdot \text{min}} \right)$$

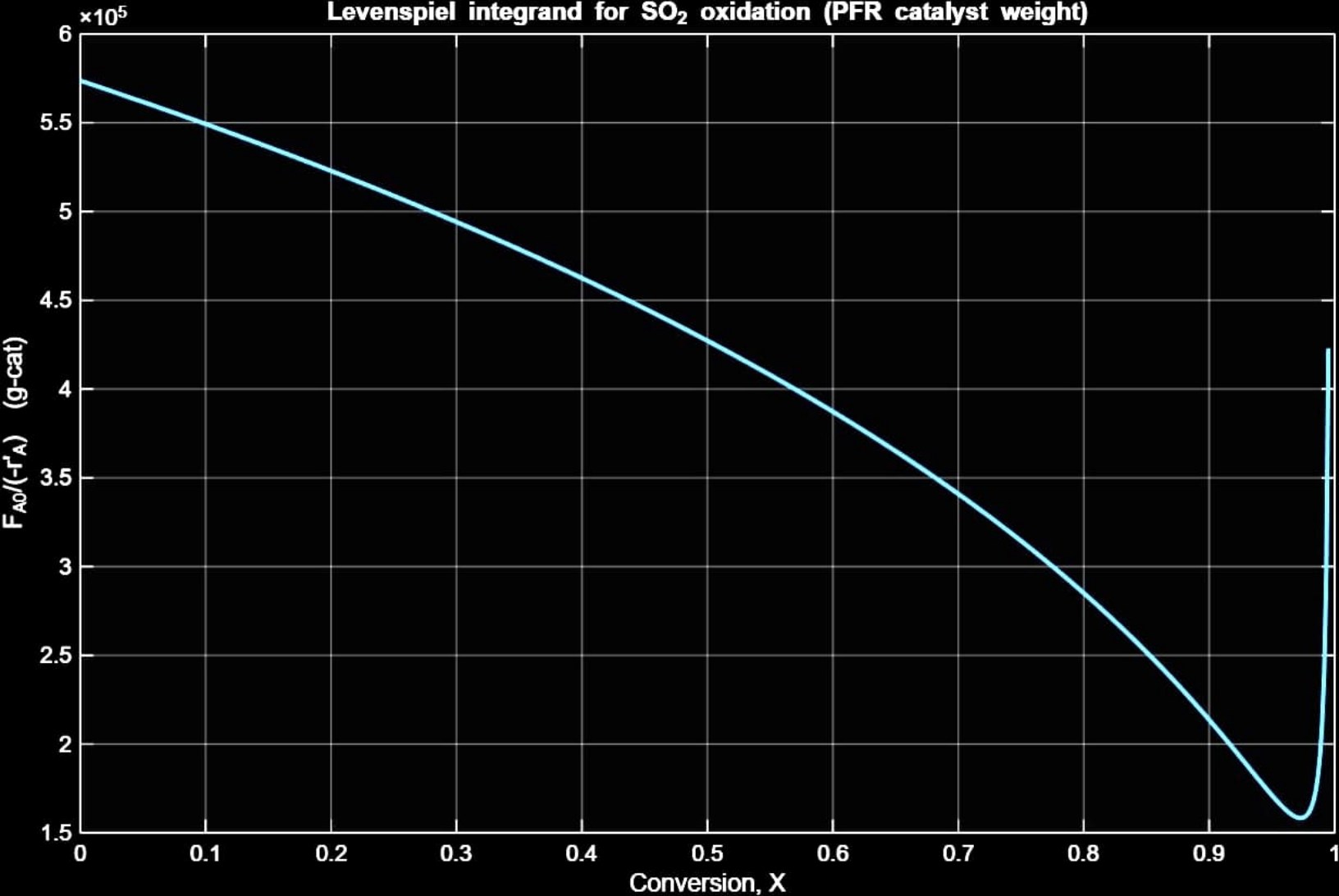
For a PFR,

$$V = \int_0^X \frac{F_{A0}}{-r_A(X)} dx$$

$F_{A0} = 3 \text{ mol mol}^{-1} \text{ s}^{-1} \leftarrow X = 0.40$

$$V_{\text{PFR}} = \int_0^{0.4} \frac{3}{-r_A(X)} dx = 69 \text{ dm}^3$$

③ SO_2 oxidation over catalyst



Problem 2



②

$$N_A = N_{A0}(1-X), \quad N_B = N_{A0}(1-X), \quad N_C = 2N_{A0}X$$

$$N_T = N_{T0}(1+3X), \quad N_{T0} = 2N_{A0}$$

Now using

$$PV = nRT, \quad V = 0.1 P$$

$$n = N_T$$

$$0.1 P^2 = nRT$$

$$V = 0.1 P$$

$$= 0.1 \sqrt{10 nRT} \propto \sqrt{n}$$

$$\Rightarrow \frac{V}{V_0} = \sqrt{\frac{N_T}{N_{T0}}} = \sqrt{1+3X}$$

Hence

$$C_A = \frac{N_A}{V} = \frac{N_{A0}}{V_0} \frac{(1-X)}{\sqrt{1+3X}} = C_{A0} \frac{(1-X)}{\sqrt{1+3X}}$$

$$\leftarrow \text{Similarly } C_B = C_{A0} \frac{(1-X)}{\sqrt{1+3X}}$$

$$\therefore C_{T0} = \frac{P_0}{RT} = \frac{1.5}{0.73 \times 600} = 3.4247 \times 10^{-3} \frac{\text{lb mol}}{\text{ft}^3}$$

$$\text{so, } C_{A0} = C_{B0} = y_{A0} C_{T0} = \frac{1}{2} C_{T0} = 1.7123 \times 10^{-3} \frac{\text{lb mol}}{\text{ft}^3}$$

$$\therefore -r_A = k_1 C_A^2 C_B = k_1 C_{A0}^3 \frac{(1-x)^3}{(1+3x)^{3/2}}$$

$$= 5.03 \times 10^{-9} \frac{(1-x)^3}{(1+3x)^{3/2}} \frac{\text{lb mol}}{\text{ft}^3 \cdot \text{s}}$$

①

$$V = 0.20 \text{ ft}^3$$

$$V/V_0 = \sqrt{1+3x}, \quad \sqrt{1+3x} = \frac{V}{V_0} = \frac{0.20}{0.15} = \frac{4}{3}$$

$$\Rightarrow 1+3x = \frac{16}{9} \Rightarrow \boxed{x = 0.259}$$

$$-r_A = 5.03 \times 10^{-9} \frac{(1-0.259)^3}{(1+3(0.259))^{3/2}} = 8.63 \times 10^{-10} \frac{\text{lb mol}}{\text{ft}^3 \cdot \text{s}}$$